

Generated Question Paper

Q1. (iii) Ancestors of monkey flower plants had a much smaller leaf area. Explain how the monkey flower plants have developed a larger leaf area over time. [5 marks]

Q2. (b) Lugworms live in sand on beaches and feed on diatoms, which require ammonium ions. Researchers tested effects of microplastics on lugworms; see Table 3.1. Explain and explain why researchers concluded that non-biodegradable microplastics are most harmful to lugworms using Table 3.1. [4 marks]

Q3. (d) Scientists had higher breathing rates in the 550 ppm CO₂ plot. [2 marks]

Q4. State two advantages of using a pyramid of energy rather than a pyramid of biomass to represent a food chain. [2 marks]

Q5. (a) (i) It is estimated that microplastics make up 5 Calculate the mass of microplastics used on one day in 2013 in these countries. Show working and units. [3 marks]

Q6. (c) Outline the reflex arc pathway in response to shining bright light into the eye. [3 marks]

Q7. (c) State the names of molecules converted into ammonium ions in the nitrogen cycle. [1 marks]

Q8. (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q9. (c) Explain why plants store starch in potato tubers. [2 marks]

Q10. (a) State the hormone released from the testes. [1 marks]

Q11. (ii) Enzymes are proteins. State the names of the chemical elements found in all proteins. [2 marks]

Q12. A: Blood vessel C. Blood vessel C transports blood from the to the . [2 marks]

Q13. (a) Describe the possible causes of increased atmospheric carbon dioxide. [3 marks]

Q14. (b) Complete Fig. 5.1 by stating two other organs that belong to two organ systems. begincenter
endcenter noindent (c) Fig. 5.2 shows photomicrograph of part of mammalian testis. begincenter
endcenter Explain why meiosis is necessary in the testes. [3 marks]

Q15. (a) Define the term 'species'. [2 marks]

Q16. (iii) State the name of the gas that forms to create the gas bubbles in the dough. [1 marks]

Q17. (ii) State the name of the process occurring at step 3 that causes gas bubbles to form in the dough. [1 marks]

Q18. (f) Describe what happens to a fertilised egg cell before implantation in the uterus. [3 marks]

Q19. (d) State the name of the process converting ammonium ions into nitrate ions. [1 marks]

Q20. (a) The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. (i) State what is meant by the term sustainable resource. [2 marks]

Q21. (ii) State a conclusion for the results shown in Table

[1 marks]